Delivery Reliability Monitoring

for an Online Marketplace

Business Requirements Document

Olist

Case study based on the Olist Brazilian E-Commerce Public Dataset

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Project sponsor	Head of Operations (assumed role)
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Practice case study using public data. Stakeholder roles and targets are assumed.

Document control

Item	Detail
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Version history

Version	Date	Revised by	Reason for change
1.0	7 October 2026	Bimal K Augustine	Initial draft for review
1.1	22 October 2026	Bimal K Augustine	Added document control, version history and approval block. Added a rationale column to the business requirements. Corrected risk and assumption numbering.
1.2	25 October 2026	Bimal K Augustine	Final review. Simplified the wording, renumbered the assumptions and marked the document final.

Document approvals

The approvers are assumed roles, because this is a practice case study. In a real project, any change after approval would be reviewed for its impact before it is accepted.

Approver	Project role	Signature / electronic approval	Date
Head of Operations	Sponsor	Pending (simulated)	
Logistics Partner Manager	Reviewer	Pending (simulated)	
Customer Experience Manager	Reviewer	Pending (simulated)	
Data and BI Team	Reviewer	Pending (simulated)	

1. Executive Summary

The marketplace in this dataset received 99,441 orders between September 2016 and October 2018, and 97.0% of them reached delivered status. Of the 96,470 delivered orders that have a delivery date, 7,826 (8.1%) arrived after the delivery date promised to the customer at the time of purchase. Late orders get much worse reviews. They average a review score of 2.57 out of 5, against 4.29 for on-time orders, and 54.1% of reviewed late orders receive 1 or 2 stars, against 9.2% for on-time orders. Lateness is also uneven: the monthly late rate peaked in November 2017, February 2018 and March 2018.

Operations has no single view of delivery performance that everyone trusts. This document sets out what the business needs from a monitoring solution: an analysis and an interactive dashboard that show how often, where and when orders are late, and how much that hurts customer satisfaction.

2. Business Problem

Problem statement. The marketplace delivers most orders, but the 8.1% that arrive after the promised date get much worse reviews. Late deliveries cluster in particular months, and operations has no consistent way to monitor delivery performance or see where it fails.

Baseline evidence (calculated from the source data):

Measure	Baseline	Basis
Orders in dataset	99,441	All orders
Orders delivered	96,478 (97.0%)	All orders
Delivered orders with a delivery date	96,470	Delivery base used for all delivery KPIs
Late orders	7,826 (8.1%)	Delivered after the promised date
Average / median delivery time	12.56 / 10.22 days	Purchase to delivery
90th percentile delivery time	23.1 days	Purchase to delivery
Average promised delivery time	23.74 days	Purchase to promised date
Orders more than 3 days late	5,164 (5.4%)	Share of the delivery base
Average review score, late vs on-time	2.57 vs 4.29	Reviewed orders
Reviewed orders scoring 1 or 2 stars, late vs on-time	54.1% vs 9.2%	Reviewed orders

Why it matters. Review scores fall sharply once an order is more than about 3 days late: orders 3 to 7 days late average 2.32 stars, against 4.29 on time. This is an association in the data, not proven cause (see assumption A-05).

3. Business Objectives

ID	Objective	Measure of success
O1	Establish a trusted baseline of delivery performance	Late rate, delivery time and review score reported with agreed definitions
O2	Identify where and when late deliveries concentrate	Late rate shown by month, customer state, seller and product category
O3	Quantify the satisfaction impact of late delivery	Review score shown by delay bucket (on time, 1-3, 3-7, 7-14, over 14 days late)
O4	Give operations a dashboard to monitor delivery performance	Interactive dashboard answering requirements BR-01 to BR-10
O5	Support improvement decisions with evidence	Recommendations that each cite supporting data

Example targets (my assumptions; in a real project these would be agreed with stakeholders): reduce the late rate from 8.1% towards 5% or lower, and reduce the share of orders more than 3 days late from 5.4% towards 3% or lower.

4. Scope

In scope

- Delivery performance of delivered orders: on-time rate, delivery time, delay severity
- Breakdown by month, customer state, seller state, seller and product category
- Relationship between delivery delay and customer review score
- Commercial context: order count and item revenue
- An interactive dashboard, supporting analysis and written recommendations

Out of scope

- Profitability and margin (the data contains no cost information)
- Forecasting or predictive modelling
- Changes to carrier contracts or the checkout promise algorithm (recommendations only)
- Repeat-purchase and retention analysis (only 3.1% of customers ordered more than once)
- Text analysis of Portuguese review comments
- Building or changing any operational system

5. Stakeholders

The stakeholder roles are assumed, because this is a practice case study.

Stakeholder	Interest	Influence	What they need
Head of Operations (sponsor)	Reduce late deliveries and complaints	High	Clear baseline, where and when problems occur, recommendations
Logistics Partner Manager	Carrier and delivery performance	High	Delay by region and over time
Seller Management Lead	Seller quality and compliance	Medium	Seller-level late rates with fair volume thresholds
Customer Experience Manager	Customer satisfaction	Medium	Link between delays and review scores
Commercial and Finance Analyst	Revenue figures	Medium	Consistent revenue definition
Data and BI Team	Data quality and dashboard upkeep	Medium	Documented definitions and data issues
Customers and sellers (indirect)	Reliable delivery	Low	Benefit from improvements, not involved in the project

The RACI matrix and process maps are in the Process Analysis document.

6. Business Requirements

Priority uses MoSCoW: Must, Should, Could. Each requirement links to an objective (section 3) and states why the business needs it.

ID	The business needs to...	Priority	Objective	Rationale
BR-01	Measure on-time and late delivery against the date promised at purchase	Must	O1	Late delivery is the core problem; every other figure depends on one agreed definition (BRU-01)
BR-02	Measure delivery time (average, median, 90th percentile) from purchase to delivery	Must	O1	Shows actual delivery speed next to the promise, because the promised date looks generous (R-02)
BR-03	See delivery performance by month to find trends and peaks	Must	O2	Lateness is uneven over time, with peaks in November 2017, February 2018 and March 2018
BR-04	See delivery performance by customer state and by same-state versus cross-state shipping	Must	O2	The Logistics Partner Manager needs delay by region to act on carrier performance
BR-05	See delivery performance by seller, ignoring sellers with too few orders to judge fairly	Should	O2	Seller Management needs seller-level late rates, and small volumes would misjudge sellers (BRU-09)
BR-06	See how review score changes with delivery delay, including severity buckets	Must	O3	Shows how much satisfaction is lost and at what point the damage starts (see section 2)
BR-07	See delivery performance by product category	Should	O2	Shows whether lateness concentrates in particular kinds of product
BR-08	See order volume and item revenue alongside delivery performance	Should	O4	Gives commercial context so late rates can be read against business size

ID	The business needs to...	Priority	Objective	Rationale
BR-09	Filter all views by date range, state and category	Must	O4	Lets each person look at just their own area
BR-10	See metric definitions and known data limitations inside the report	Must	O1, O4	Stops users misreading figures, for example reading the review link as proven cause (A-05)

Detailed functional requirements, non-functional requirements, user stories and acceptance criteria are documented in the Requirements Specification.

7. Business Rules

ID	Rule
BRU-01	An order is late when the delivery date to the customer is after the estimated delivery date that was shown to the customer at purchase
BRU-02	Delivery KPIs use orders with status delivered and a delivery date (96,470 orders)
BRU-03	Delay severity buckets: on time, 1-3, 3-7, 7-14 and over 14 days late
BRU-04	Where an order has several reviews, the first review is used
BRU-05	Customers are counted with customer_unique_id, not customer_id
BRU-06	Revenue is the sum of item price. Freight is reported separately. Canceled and unavailable orders are excluded from revenue
BRU-07	Tables are totalled per order before they are joined, to avoid repeated rows and inflated revenue
BRU-08	Monthly trends cover January 2017 to August 2018, because other months have too few orders. Headline KPIs cover all 96,470 delivered orders
BRU-09	Seller rankings include only sellers with at least 30 delivered orders (my assumed threshold)
BRU-10	An order is same-state when the customer state equals the state of the seller of the order's first item. Orders with several sellers are classified by that first item

8. Assumptions

The main assumptions are:

ID	Assumption
A-01	The currency is Brazilian real (BRL); the documentation does not state it
A-02	Order status meanings are inferred from the data, for example that unavailable orders could not be supplied
A-03	The timezone of the timestamps is not stated; day-level analysis limits the effect
A-04	customer_unique_id identifies a person; customer_id is created per order
A-05	Review scores reflect the delivery experience, so the link to lateness is an association, not proven cause
A-06	Stakeholder roles and improvement targets are assumed, because this is a practice case study

9. Constraints

- Only the supplied dataset is available, covering September 2016 to October 2018
- The data contains no cost or margin information
- The data is historical, so the dashboard is built on a static extract and not a live feed
- There are no real stakeholders to interview, so the requirements are my proposals, not gathered from people
- Tools are limited to Excel (with Power Query), SQL and Power BI

10. Risks and Limitations

ID	Risk or limitation	Handling
R-01	Late delivery and low ratings may both be caused by other factors	Describe it as a link, not a cause
R-02	The promised date looks generous (orders are promised 23.74 days ahead on average but arrive in 12.56 days on average), so the on-time rate may flatter performance	State this next to the late rate
R-03	Only customers who reviewed are represented in satisfaction figures	Show the number of reviewed orders
R-04	Orders from the final months may be undelivered when the data ends, which can make recent months look better	Mark recent months with a caution
R-05	Months before January 2017 and from September 2018 are too sparse to compare	Excluded from trend analysis
R-06	Joining tables can repeat rows and inflate revenue	Business rule BRU-07

11. Success Criteria

- Each business requirement BR-01 to BR-10 is met in the Excel workbook, SQL queries or Power BI dashboard, or is listed as a known gap
- Dashboard figures reconcile with the Excel and SQL analysis
- Metric definitions and data limitations are visible in the report
- Each recommendation cites supporting evidence
- Requirements trace to analysis, dashboard views and insights in the traceability matrix

12. Dependencies

- Agreement on the definitions in section 7
- Access to the source data extract and its documentation
- Review of the assumed thresholds and targets by stakeholders

Appendix A. Glossary

Term	Meaning
Promised date	The estimated delivery date shown to the customer at purchase
Delivery days	Days from purchase to delivery to the customer
Late order	Delivered after the promised date
Days late	Delivery date minus promised date; negative means early
Review score	Customer rating from 1 to 5
Seller	The merchant who sells and ships the item
Customer (person)	Identified by customer_unique_id

Appendix B. Data Attribution

Brazilian E-Commerce Public Dataset by Olist, published on Kaggle. Licence: CC BY-NC-SA 4.0. Used for educational and portfolio purposes only.

Appendix C. How this document was prepared

This Business Requirements Document was prepared with the assistance of Claude, an AI assistant from Anthropic. Claude made the documentation process smoother by suggesting the industry-standard structure, helping with the wording, and checking that the numbers match across the Excel, SQL and Power BI work. It also answered my questions and pointed out my mistakes. The data analysis in Excel, SQL and Power BI, the dashboard and the diagrams were done by me. Claude helped me with some of the SQL queries, which I ran and checked against my Excel results. The stakeholders, roles and targets are assumed, because this is a practice case study.